



TableRAG: Retrieval-Augmented Generation for Heterogeneous Document Reasoning

Presenting a framework for reasoning over tables and text together

Zheng et al., 2025 | arXiv preprint

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The Problem: Structure Loss in Real Documents

Real-World Challenge

Documents contain both text and tables with critical structured data. Traditional LLMs struggle when tables lose their row-column structure through flattening.

Core Issue

How do we reason over tables + text together without flattening structure?

Dataset & Task Setup



Heterogeneous Documents

Text passages combined with tables containing rows and columns of structured data



Question Answering

Locating relevant tables, selecting specific rows, and combining with text context



Benchmark Datasets

Public document QA datasets with tables requiring numerical and factual reasoning

Works Cited

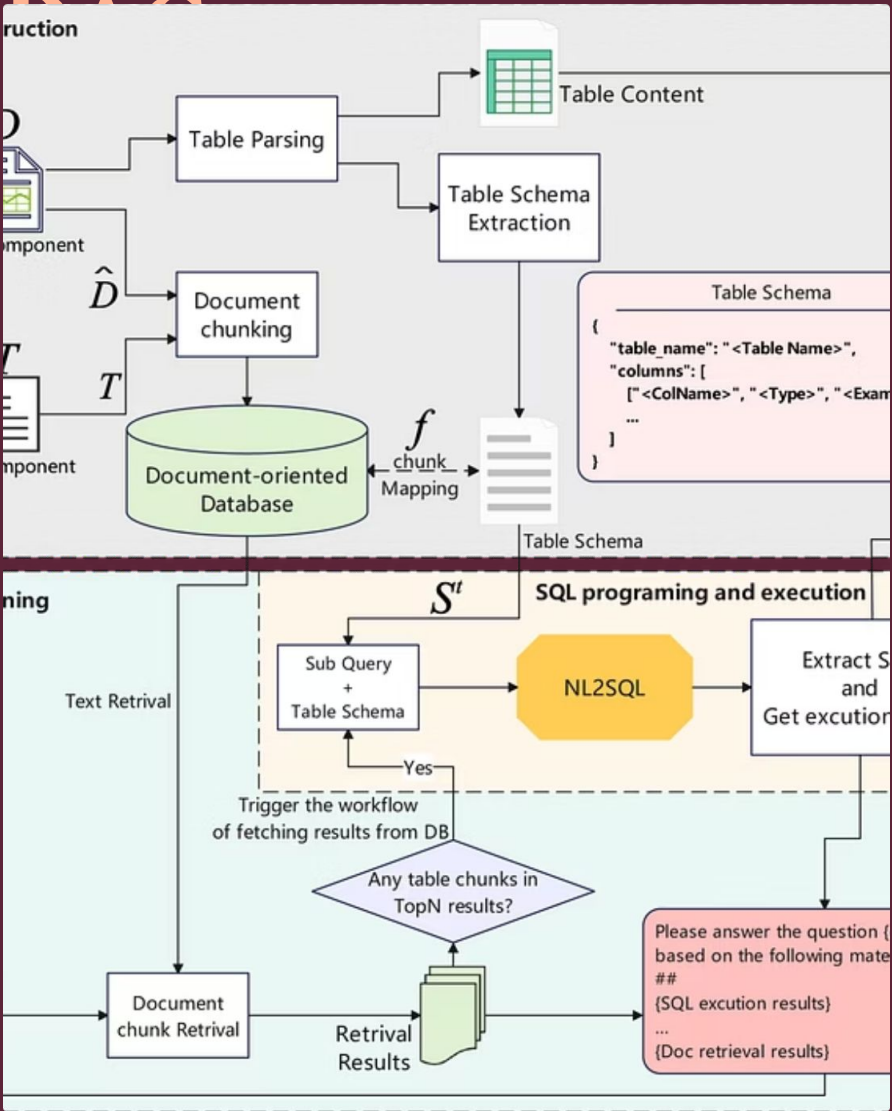
Zheng, et al. "TableRAG: A Retrieval Augmented Generation Framework for Heterogeneous Document Reasoning." arXiv, 2025.

arXiv:2411.02059.

arXiv:2504.01346.

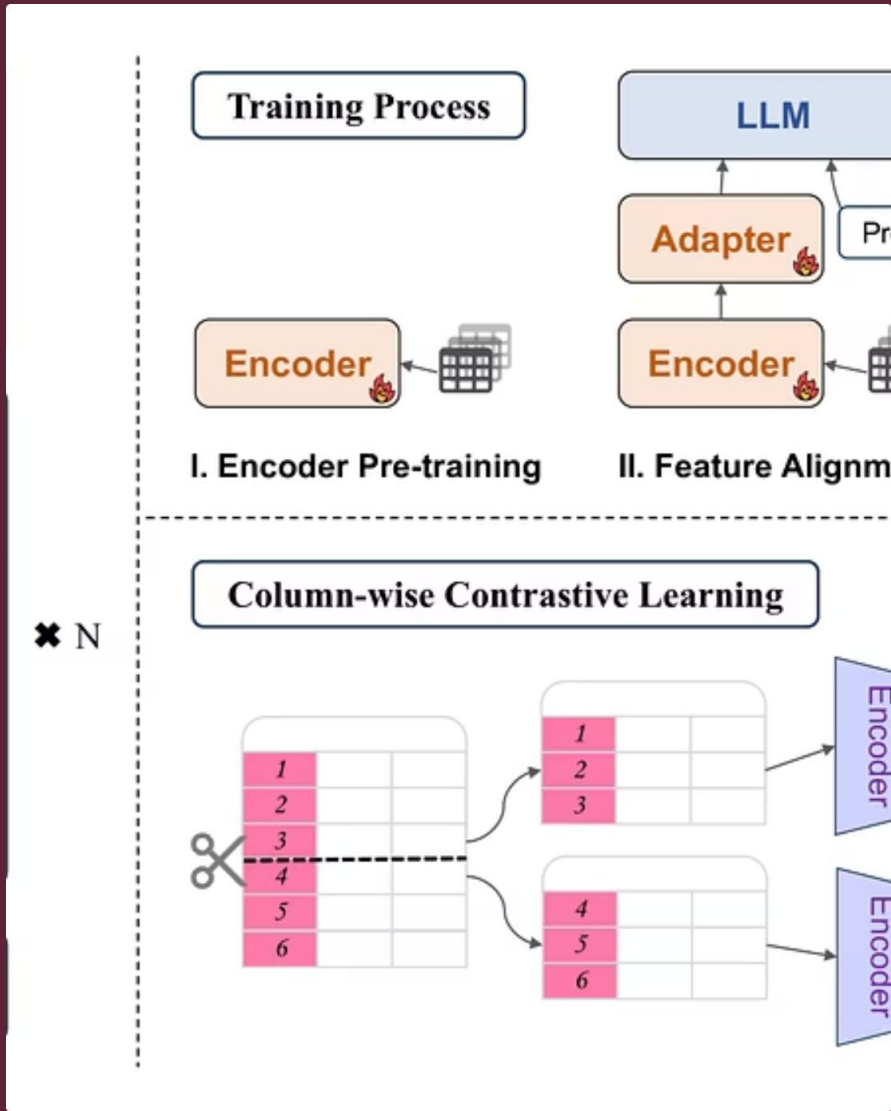
arXiv:2410.04739.

Key Works in Table-Aware



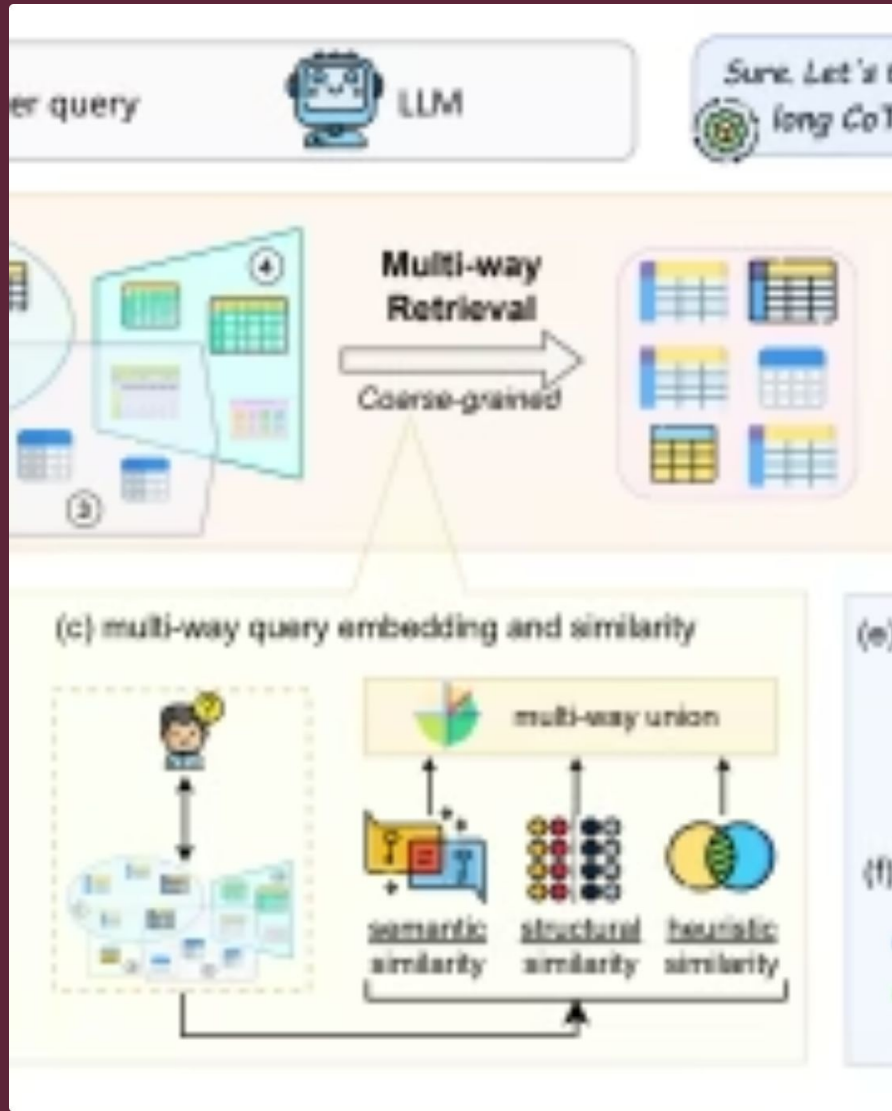
TableRAG (Primary)

Keeps table structure during retrieval and reasoning at document, table, and row levels



Paper A (arXiv:2411.02059)

Table-aware retrieval with limitations at generation stage



Paper B (arXiv:2504.01346)

Converts tables into structured prompts, heavily prompt-dependent



Why TableRAG Stands Out

1

First-Class Table Objects

Tables treated as structured entities, not flattened text

2

Hierarchical Retrieval

Document → table → row level selection

3

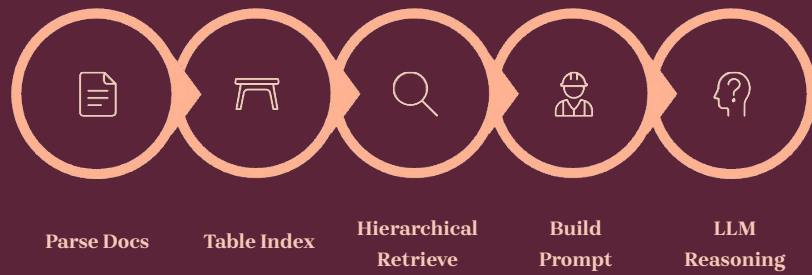
Structured Generation

LLM reasoning grounded in retrieved structured evidence

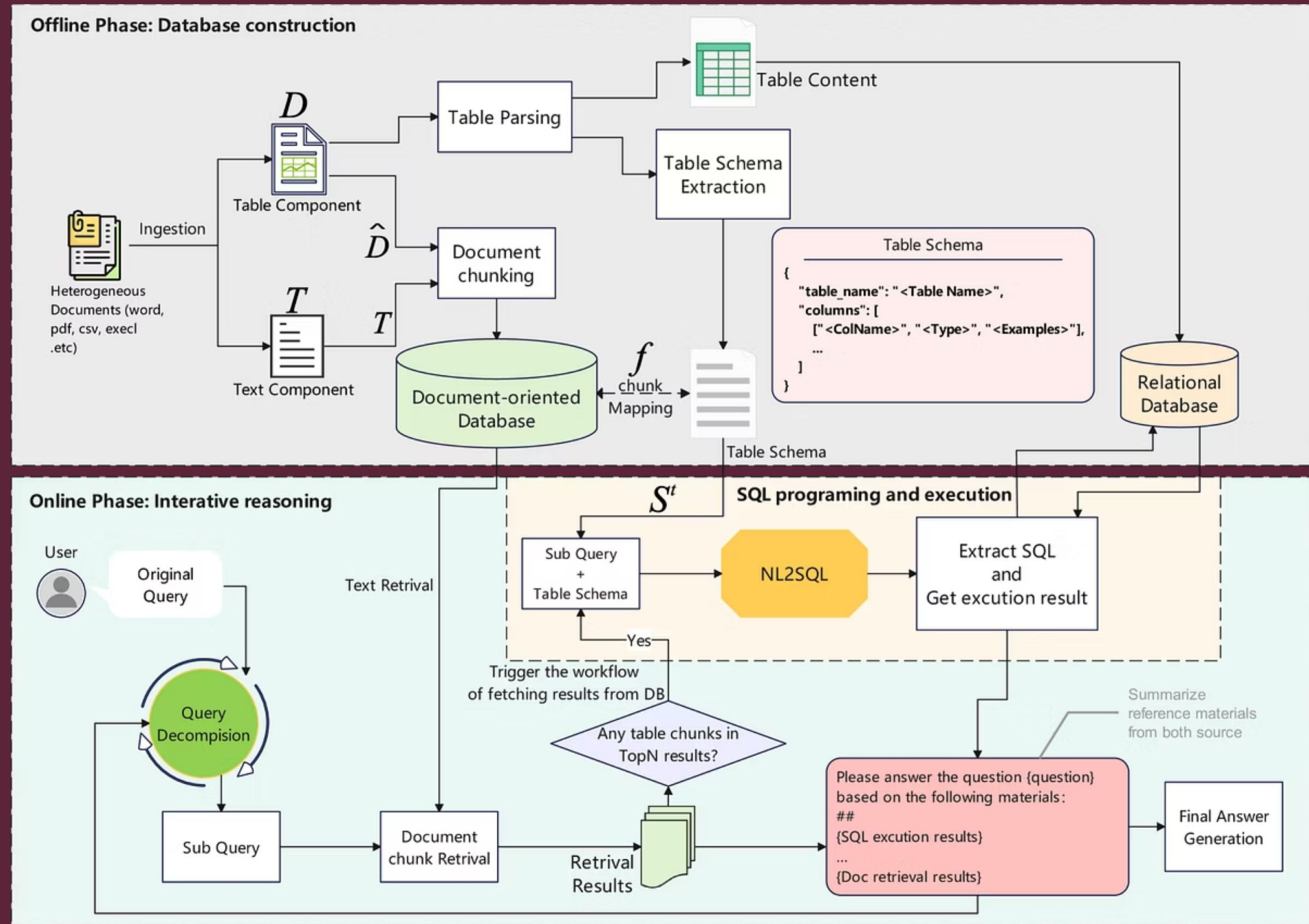
Clean separation of retrieval and reasoning makes it reproducible and extensible

TableRAG

Method: Core Architecture



Key innovation: Retrieve only the rows that matter, not entire tables





Retrieval Strategy: Multi-Level Indexing

Separate Embeddings

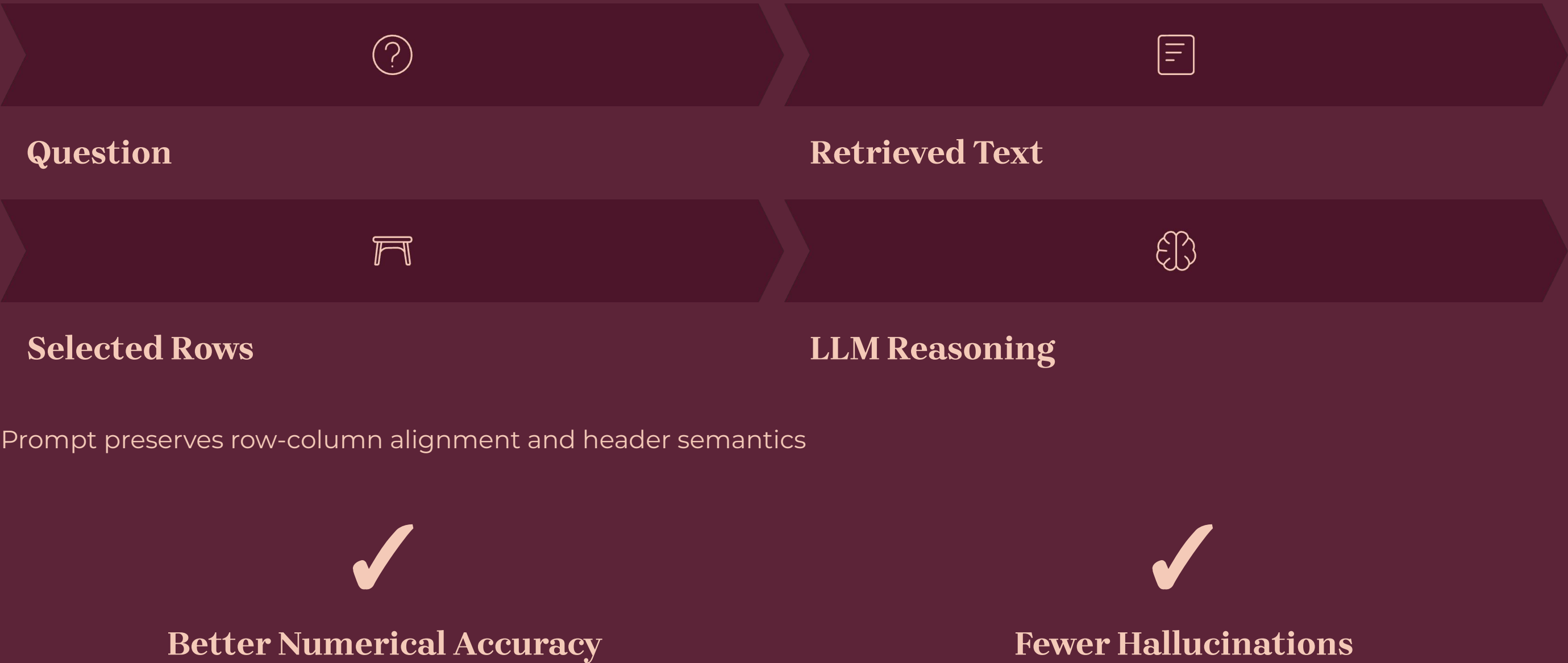
- Text chunks
- Tables
- Table rows

Progressive Retrieval

Question → relevant tables → relevant rows

Reduces noise before generation

Generation & Reasoning Process



Reproduction Plan & Timeline



Expected Learning Outcomes



Practical Limits

Understanding RAG with structured data



Design Trade-offs

Retrieval granularity decisions



Breaking Assumptions

How tables challenge standard NLP assumptions



Prompt Grounding

Factual accuracy through structured evidence

